

Section 3.5 Implicit Differentiation Part 2

$$\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

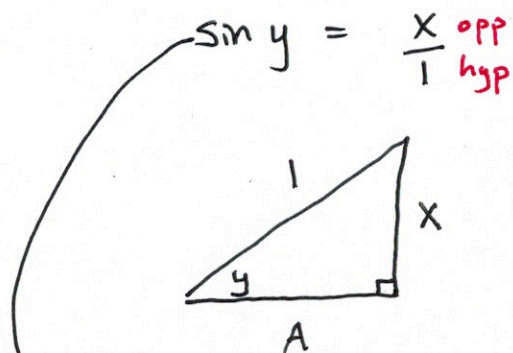
$$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx} (\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} (\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

Proof $y = \sin^{-1} x$



$$A^2 + x^2 = 1^2$$

$$A^2 = 1 - x^2$$

$$A = \sqrt{1-x^2}$$

$$\cos y \cdot y' = 1$$

$$y' = \frac{1}{\cos y}$$

$$\text{where } \cos y = \frac{\sqrt{1-x^2}}{1} \therefore y' = \frac{1}{\sqrt{1-x^2}}$$

Ex ① $y = \tan^{-1}(\sqrt{x})$ Find $\frac{dy}{dx}$

$$y' = \frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}}$$

$$y' = \frac{1}{(1+x) \cdot 2\sqrt{x}}$$

Ex ② $y = \sqrt{\sin^{-1}x}$

$$y' = \frac{1}{2\sqrt{\sin^{-1}x}} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\boxed{\frac{d}{dx}(\ln x) = \frac{1}{x}}$$

Ex ③ $y = \ln x$

$$e^y = x$$

$$e^y \cdot y' = 1$$

$$y' = \frac{1}{e^y}$$

$$y' = \frac{1}{x}$$

Ex ④ $y = 2^x$

$$\ln y = \ln 2^x$$

$$\ln y = x \cdot \ln 2$$

$$\frac{1}{y} \cdot y' = 1 \cdot \ln 2$$

$$y' = y \cdot \ln 2$$

$$y' = 2^x \cdot \ln 2$$

$$\frac{d}{dx}(a^x) = a^x \cdot \ln a$$

Ex ⑤ $y = (\sin^{-1} x)^2$

$$y = (\arcsin x)^2$$

$$y' = 2 \cdot \arcsin x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$y' = \frac{2 \cdot \sin^{-1} x}{\sqrt{1-x^2}}$$

Ex ⑥ $y = \ln(\sec x)$

$$y' = \frac{1}{\sec x} \cdot \sec x \cdot \tan x$$

$$\boxed{y' = \tan x}$$

Compare with ...

$$y = -1 \cdot \ln(\cos x) \rightarrow \ln(\cos x)^{-1}$$

$$= \ln\left(\frac{1}{\cos x}\right)$$

$$= \underline{\underline{\ln(\sec x)}}$$

$$\boxed{y' = \tan x}$$